where the exclusion zone will not even cross the boundary of the plant. This will probably be put into service by 2033. A third design, the Indian Gas Cooled Reactor (IGCR) for high-temperature hydrogen generation, is in the conceptual stage.

For Indian Railways, such smaller reactors may be the answer to 24/7 clean base load power, freed of the coal-based national electricity grid. Indeed, in many places, railway entities are not even waiting for an official directive—they are already clearing the decks. M.P. Singh, director (operations) of Rail Vikas Nigam Limited (RVNL), a key infrastructure arm under the ministry, says, "We now have smaller N-power reactors of 55-60 MW, and they don't take more than 24 acres of land. We are trying to tie up with the Russian company, Rosatom, who are experts at setting up these kind of reactors."

Greening from the Top

Indian Railways has also called for 2 GWe (gigawatts-electric) of nuclear power from the power ministry, offering land and guaranteed power offtake in return. DAE and NPCIL will take care of constructing, licensing and operating the reactors. Feasibility studies are on, including for possible locations near key traction substations and industrial freight routes, says a source privy to the discussions.

Money is also beginning to flow in. Union Budget 2025 has provisioned Rs 20,000 crore for SMR development, with five deployments by 2033. Indian Railway Finance Corporation (IRFC) is likely to have a role in this. "Our mandate is to secure the funding for all renewables for Railways. For nuclear energy, we will secure funding at the cheapest cost. We are doing that," says Manoj Dubey, chairman and managing director of IRFC.

The railway ministry is not the only one betting on nuclear power. The wider Nuclear Energy Mission, announced earlier this year, seeks to achieve 100 GWe of nuclear power by 2047, to help India meet its Net Zero 2070 national goal. SMRs are the thin edge of that effort, and railways could be a marquee consumer.

But the nuclear pivot would only work as part of the larger energy turnaround that's already under way. These shifts have saved the Railways more than Rs 2 lakh crore in fuel costs and logistics over the past decade. Meanwhile, carbon emissions are set to go down from 3.3 MTPA (million tonnes per annum) in 2023-24 to 1.4 MTPA by 2025-26.

A CASE FOR SMR

- **A 100% electrified Railways will need 10 GW of uninterrupted power annually by 2030**
- **Small modular reactors (SMRs) can give 24x7 clean power, require less space, can be built at faster pace**
- **Talks on with NPCIL, Russia's Rosatom. Indian Railways set to emerge as marquee user of atomic energy**
- **The indigenous BSMR-200 and SMR-55 reactors could in time provide reliable power to railways**



"N-power is clean and can be cheaper than many other sources (of power), apart from being free from disruptions"

ASHWINI VAISHNAW
Union railways minister

There is some modest success in this already. More than 1,950 railway stations now run on solar power to some extent. Round-the-clock renewable procurement has begun. And to all these changes, there is a quantifiable environmental angle: millions of tonnes of carbon dioxide emissions saved—the equivalent of planting some 1,210 million trees, according to a Railways estimate.

A Climate Story on Rails

The big picture, then, is the Indian Railways' energy strategy: renewables mixed with grid power. With India's commitment to Net Zero, transportation continues to be a key sector for decarbonisation. But with rail almost 90 per cent more carbon-efficient than road (per tonne km), it is already a greener option. Greening its own power supply closes the loop, experts say.

In June 2023, Indian Railways also signed a non-binding MoU with USAID to sustain its mission of 'Net Zero Carbon Emission by 2030'. With this, technical aid is available for clean energy planning, large scale renewables procurement, grid integration, energy efficiency in railway operations and in buildings. While it's not a direct financial commitment, the coordination can make up for deficiencies in technology and skills.

For, there are challenges ahead. The SMR plans are still on the drawing board, and even the most optimistic estimates hope for construction to start in 2031. There will be indispensable regulatory approvals and local opposition as well to navigate around. But the signal has been sent. From being an impractical fantasy even a decade ago, the idea of nuclear-powered public transit in India has finally entered the realm of reality. From huffing diesel behemoths that belch fumes to solar-powered stations that run almost silently; and now the prospect of atomic electricity. The pendulum has already started swaying from 'if' to 'when'. ■